

CONSULTANCY PROPOSAL AND FINANCIAL QUOTATION

Project: Beethoven AI Production-Ready Orchestral Score Generation with NeuralConstraints/Gecode

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Client: imec and Ghent University (IPEM, MICT), Beethoven AI project

Proposed period: 15 September 2026 1 May 2027

Workload: Part-time, approximately 20% FTE, with effort concentrated around integration, testing, and rehearsals (estimated total: 268 hours)

1. PURPOSE AND OBJECTIVE

This proposal covers the remaining consultancy work required to advance the existing NeuralConstraints/Gecode musical constraint solver from its current research/prototype stage to a production-ready and performance-stable component of the Beethoven AI live-performance ecosystem.

The objective is to deliver a dependable orchestral score-generation service that can transform selected Beethoven source material and higher-level musical instructions into playable, constraint-valid score material during rehearsal and performance workflows. The assignment includes production engineering, integration, validation, deployment, documentation, and rehearsal-driven refinement. Previously completed and university-funded foundational research, architecture, and prototype implementation are excluded from this quotation.

2. EXISTING FRAMEWORK

NeuralConstraints/Gecode is a C++17 polyphonic music-generation framework built on the Gecode constraint-programming library and inspired by the rule-based, multi-voice approach of Cluster Engine. The existing implementation provides:

- A shared solver core for command-line, test-wrapper, and native Max/MSP external interfaces.
- Explicit pitch, rhythm, rest, meter, tuplet, harmonic, and multi-voice domains.
- Local and global musical constraints, including melodic, rhythmic, metric, harmonic, voice-leading, and inter-voice rules.
- Built-in constraints and JSON-authored dynamic or wildcard rule expressions, allowing musical requirements to be changed without recompiling the solver.
- Gecode propagation and backtracking, which guarantee that every returned solution satisfies all enabled hard constraints.
- Heuristic and neural candidate ranking for Beethoven-oriented stylistic guidance. Neural scoring influences which valid material is explored first; it cannot override hard musical or technical constraints.
- Configurable search controls including timeout, solution limits, deterministic seeds, branching strategy, value ordering, and restart policy.

- Structured JSON configuration and result handling, plus score export to MusicXML and other supported interchange formats.
- A native asynchronous Max/MSP integration path and a macOS Max package deployment workflow.

The consultancy therefore develops and hardens an existing system rather than funding a new solver from first principles.

3. SCOPE OF WORK AND DELIVERABLES

3.1 Production-level music generation and orchestration

- Consolidate the existing melodic, harmonic, rhythmic, metric, and polyphonic rule system into production configurations for Beethoven AI.
- Develop and tune orchestral voice and instrument profiles, including ranges, register, texture, voice allocation, rhythmic behavior, and practical playability constraints agreed with the artistic team.
- Integrate Beethoven-derived sketch material and project-supplied harmonic or structural instructions into the solver's generation pipeline.
- Refine the interaction between hard constraints, musical heuristics, and neural candidate ranking so generated material remains both stylistically directed and constraint-valid.
- Stabilize score extraction and MusicXML/JSON output required by downstream notation, rendering, or performance components.

Deliverable: a versioned production configuration and solver build capable of generating agreed orchestral test passages from representative Beethoven AI inputs.

3.2 Runtime reliability and performance

- Profile and optimize propagation, search, rule compilation, score extraction, and repeated-solve behavior under representative orchestral workloads.
- Establish bounded runtime behavior using timeouts, cancellation, maximum solution counts, deterministic seeds, and controlled fallback behavior.
- Prevent state leakage between successive requests and provide explicit success, unsatisfiable, timeout, cancellation, and configuration-error states.
- Define and measure latency, memory, stability, and repeatability targets with the project team using agreed rehearsal scenarios.
- Add diagnostic logging and operational statistics suitable for troubleshooting without interrupting the live-performance process.

Deliverable: a benchmarked runtime build with documented operational limits, failure handling, and recovery procedures for the agreed performance setup.

3.3 Integration within the broader Beethoven AI ecosystem

- Finalize and version the JSON specification for musical domains, rules, Beethoven source references, search controls, and output metadata.
- Harden the common solver pipeline used by the CLI, wrapper, and Max/MSP external so all integration paths apply the same configuration semantics.
- Provide stable asynchronous request, status, result, error, timeout, and cancellation behavior for the host system.
- Expose agreed runtime control parameters, including seed, timeout, generation strategy, heuristic/neural settings, and output selection.
- Support integration testing with the AI agent and project components supplied by imec, IPEM, MICT, and their designated partners.

Deliverable: an integration-tested solver interface, JSON schema/examples, and versioned Max/MSP/CLI artifacts for the agreed host environment.

3.4 Testing and rehearsal support

- Maintain focused unit, rule-compilation, multi-voice, metric, export, wrapper, and end-to-end regression tests.
- Build a representative acceptance corpus covering normal generation, difficult/unsatisfiable inputs, repeated requests, timeout/cancellation, and score-export validity.
- Validate generated material with the research and artistic teams against agreed musical and technical acceptance criteria.
- Provide scheduled remote rehearsal support, diagnosis, and iterative bug fixes within the quoted hours.
- Prioritize and resolve issues discovered under realistic rehearsal timing and integration conditions.

Deliverable: test results, an issue/decision log, rehearsal-ready releases, and a final acceptance report listing resolved issues and any documented limitations.

3.5 Deployment and documentation

- Prepare reproducible macOS builds and a versioned Max package with required solver assets, example configurations, and neural weights where applicable.
- Harden package validation, smoke testing, installation, artifact management, and rollback procedures for the agreed deployment machines.
- Document installation, configuration, rule authoring, runtime controls, monitoring, troubleshooting, and recovery.
- Deliver technical handover to designated project staff and incorporate final rehearsal findings before the production-readiness deadline.

Deliverable: a deployable release package, deployment/runbook documentation, configuration reference, and technical handover session.

4. ACCEPTANCE BASIS

Detailed quantitative targets will be agreed during the September planning phase after representative orchestral inputs and the target performance hardware are available. Acceptance will be based on the following minimum principles:

- All returned scores satisfy the enabled hard musical constraints.
- The solver handles valid, invalid, unsatisfiable, timed-out, and cancelled requests without crashing or contaminating subsequent requests.
- Agreed JSON inputs and outputs remain stable across the production interfaces.
- MusicXML and other agreed outputs open correctly in the project's designated downstream software for the acceptance corpus.
- Representative scenarios meet the jointly agreed latency and stability targets on the designated rehearsal/performance hardware.
- The release can be installed, smoke-tested, operated, and rolled back by the project team using the delivered documentation.

5. EXCLUDED SCOPE

The following are not included unless separately agreed in writing:

- Foundational research and system architecture already completed and funded through university resources.
- New audience-facing or performer-facing applications, including multi-screen orchestral score viewers.
- Music-information-retrieval systems for biofeedback, gesture capture, or other sensor modalities.
- Composition or inference of complete high-level musical form beyond the structural inputs and interfaces agreed for this assignment.
- Procurement of hardware, commercial notation software, Max/MSP, or third-party licenses other than the development resources listed in the budget.
- On-site travel, accommodation, per diem, freight, and other mobility costs.
- Major requirements introduced after scope and acceptance criteria are locked.
- Long-term operations, feature development, or maintenance after 1 May 2027.

Any excluded or materially changed work requires written approval and a separate estimate. Travel requested by the client will be quoted or reimbursed separately and will not be undertaken without prior written approval.

6. TIME AND EFFORT PLAN

September 2026 24 h Technical planning, acceptance criteria, roadmap lock

October 2026	32 h	Production music-generation implementation
November 2026	32 h	Orchestration rules, integration, runtime controls
December 2026	30 h	Test harness, performance, and output-quality tuning
January 2027	28 h	Reliability hardening and integration fixes
February 2027	34 h	Staging, deployment workflow, rehearsal adjustments
March 2027	44 h	Acceptance testing, rehearsals, issue resolution
April 2027	44 h	Final rehearsals, documentation, handover, go-live prep

Total	268 h	

The monthly allocation is indicative. Hours may be reallocated across months by mutual agreement to follow integration and rehearsal needs without changing the maximum contract value.

7. FINANCIAL QUOTATION (EUR, TAX INCLUDED)

Professional consultancy fee

(estimated effort: 268 hours)..... EUR 20,000

TOTAL AMOUNT PAYABLE, TAX INCLUDED..... EUR 20,000

The total amount payable under this quotation will not exceed EUR 20,000, inclusive of any applicable tax. No tax or project-resource charge will be added to this amount. The applicable VAT treatment, if any, will be determined by the parties' finance departments and reflected on the invoices accordingly.

The client will provide the project resources required for the assignment, including compute, storage, development infrastructure, and relevant software access. The 268-hour allocation is an effort estimate used for planning. The fee will be invoiced against the milestones below. Work outside the agreed scope requires written approval.

8. PROPOSED INVOICING SCHEDULE

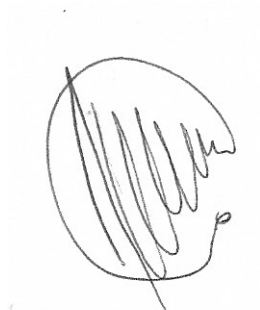
- 20% (EUR 4,000, tax included) following kickoff and interface agreement.
- 25% (EUR 5,000, tax included) following the first compiled integration release.
- 25% (EUR 5,000, tax included) following multi-block generation review.
- 20% (EUR 4,000, tax included) following the final compiled core release.
- 10% (EUR 2,000, tax included) following documentation, handover, and final acceptance.

Payment terms and tax treatment will follow the final consultancy agreement. In all cases, the combined invoiced amount will not exceed EUR 20,000.

9. CLIENT DEPENDENCIES AND WORKING ASSUMPTIONS

The client will provide timely access to representative Beethoven materials, musical specifications, target integration APIs, target hardware/software, rehearsal schedules, and designated technical and artistic decision-makers. Third-party materials and models supplied by the client are assumed to be cleared for the intended research, development, rehearsal, and performance use.

Work will normally be performed remotely. Dates depend on timely feedback and availability of the client-controlled integration environment. Source-code ownership, licensing, confidentiality, publication, data protection, liability, and rights to pre-existing versus newly commissioned intellectual property will be governed by the final written consultancy agreement.

A handwritten signature in black ink, consisting of a series of loops and a final flourish, enclosed within a circular outline.

Dr. Juan S. Vassallo

Bergen, 8 September 2026